

Decision Making Under Uncertainty | HW2

Mahir Nagersheth (mnagersh)

Canvas Excel uploader: mnagersh

Submission file: DMU_HW2_mnagersh

Page guide: Problem 1, pp. 1-4 and 16; Problem 2, p. 5; Problem 3, pp. 6-12 and 17-20; Problem 4, pp. 13-15 and 21. Pages 16-21 contain actual Excel Solver Parameters screenshots.

Problem 1(a). Linear program

Let x_j be the number of meals of Dish j prepared, $j = 1, 2, 3$. Parameters are unit costs $c = (4, 1, 3)$ dollars, cooking times $t = (3, 2, 2)$ minutes, and storage requirements $a = (0.10, 0.20, 0.15)$ shelves. Capacity is 180 minutes and 8 shelves; demand is 60 meals; each dish has a minimum of 10 meals.

$$\text{Minimize } C = 4x_1 + x_2 + 3x_3$$

subject to:

$$3x_1 + 2x_2 + 2x_3 \leq 180 \quad (\text{cooking time})$$

$$0.10x_1 + 0.20x_2 + 0.15x_3 \leq 8 \quad (\text{storage})$$

$$x_1 + x_2 + x_3 \geq 60 \quad (\text{demand})$$

$$x_j \geq 10, j = 1, 2, 3 \quad (\text{variety}).$$

Use continuous variables for the LP and its sensitivity report. The optimum is already integral, so it is also a valid whole-meal plan.

Problem 1(b). Excel Solver solution

Prepare 35 Dish 1 meals, 15 Dish 2 meals, and 10 Dish 3 meals, at a minimum cost of \$185. This serves 60 students, uses 155 minutes and all 8 shelves, and meets every variety requirement.

Input / variable	Dish 1	Dish 2	Dish 3	
Unit cost		4	1	3
Cooking minutes		3	2	2
Shelf space		0.1	0.2	0.15
Minimum meals		10	10	10
Meals to prepare				
	35		15	10
Total cost	\$185.00			

Constraint	Used / LHS	Relation	Limit / RHS	Slack
Cooking time	155	$\leq$	180	25
Shelf storage	8	$\leq$	8	0
Total demand	60	$\geq$	60	0
Minimum Dish 1	35	$\geq$	10	25
Minimum Dish 2	15	$\geq$	10	5
Minimum Dish 3	10	$\geq$	10	0

Excel worksheet image: P1 Lunch!A4:E20. Decisions B10:D10; costs B5:D5; time B6:D6; storage B7:D7; minima B8:D8; objective B12; constraint limits D15:D20.

Problem 1(c-e) | Sensitivity and binding constraints

(c) Sensitivity report

The following sensitivity values were derived from the optimal LP basis and checked against the Excel solution. This is an **analytical sensitivity report**; the automated native Excel report command did not create a report sheet. All ranges below vary one input at a time.

Variable	Value	Cost	Allowable increase	Allowable decrease
Dish 1	35	4	1	3
Dish 2	15	1	1	Unlimited
Dish 3	10	3	Unlimited	0.5

Constraint	LHS	RHS	Shadow price	Allow. increase	Allow. decrease
Cooking time	155	180	0	Unlimited	25
Storage	8	8	-30	2.5	0.5
Demand	60	60	7	5	12.5
Dish 1 minimum	35	10	0	25	Unlimited
Dish 2 minimum	15	10	0	5	Unlimited
Dish 3 minimum	10	10	0.5	10	10

With the minimum constraints represented explicitly, the variable reduced costs are zero and the Dish 3 minimum has shadow price 0.5. If minimum quantities are represented as variable bounds, the Dish 3 reduced cost is 0.5 instead. These are equivalent representations.

(d) Binding constraints

Storage, demand, and the Dish 3 minimum bind: their left-hand sides equal their right-hand sides. All storage is used, exactly 60 meals are prepared, and exactly 10 Dish 3 meals are made. Cooking time has 25 minutes of slack; Dish 1 and Dish 2 minima have slack of 25 and 5 meals.

(e) Shadow prices in context

Storage: -\$30 per shelf. An additional shelf lowers minimum cost by \$30 while storage capacity is between **7.5 and 10.5 shelves**. For example, 9 shelves would give cost $185 - 30 = \$155$. This does not value extra cooking time or a simultaneous demand change.

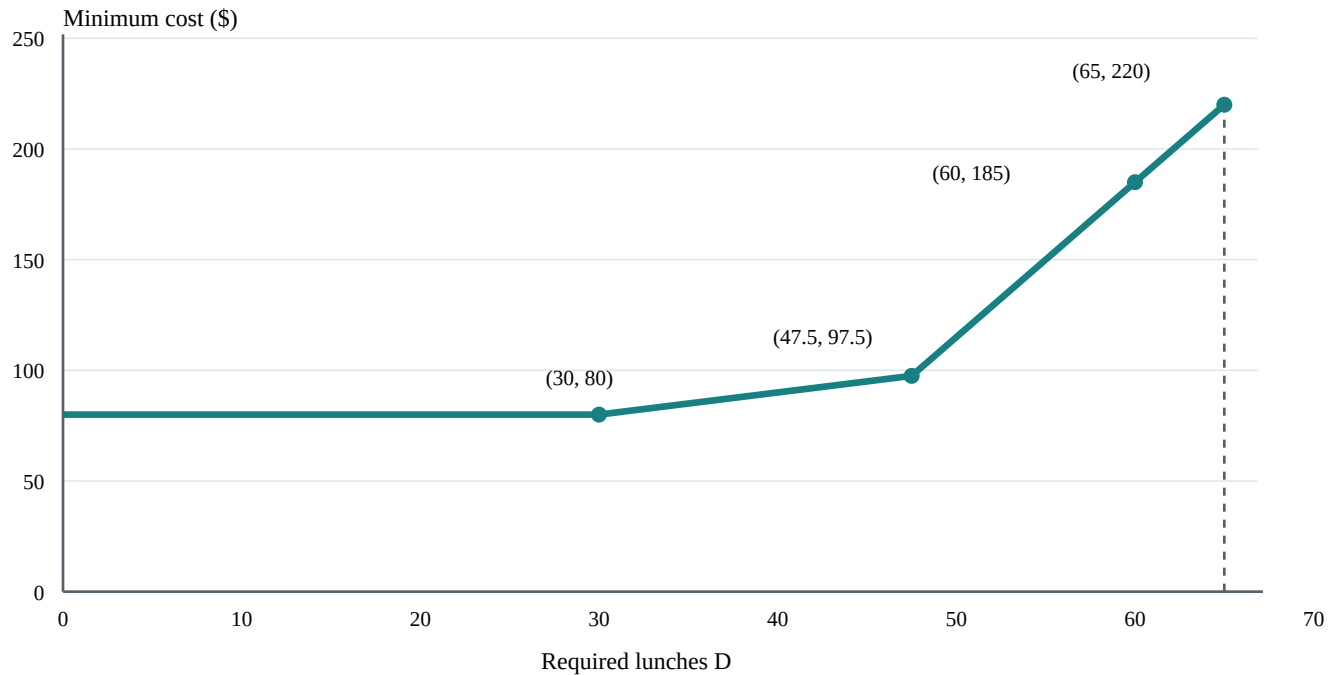
Demand: +\$7 per meal. An additional required lunch raises minimum cost by \$7 while demand is between **47.5 and 65 meals**. For integer demand, the range is 48 through 65. For example, 61 meals cost \$192.

Dish 3 minimum: +\$0.50 per meal. Increasing the required Dish 3 minimum by one raises optimal cost by \$0.50 while this minimum is between **0 and 20 meals**. For example, a minimum of 11 produces cost \$185.50 in the continuous LP.

At range endpoints a different basis can become optimal; these marginal interpretations need not continue beyond the endpoints.

Problem 1(f) | Cost as lunch demand changes

Let D be the required number of lunches, keeping all other parameters fixed. Since all unit costs are positive, the school prepares $\max(D, 30)$ meals whenever the model is feasible. The 30-meal floor comes from the three dish minima.



Continuous LP value function. The point $D = 60$ is the assigned case; demand above 65 is infeasible.

Demand interval	Optimal (Dish 1, Dish 2, Dish 3)	Minimum cost
$0 \leq D \leq 30$	(10, 10, 10)	\$80
$30 \leq D \leq 47.5$	(10, $D - 20$, 10)	$\$D + 50$
$47.5 \leq D \leq 65$	($2D - 85$, $75 - D$, 10)	$\$7D - 235$
$D > 65$	Infeasible	No finite cost

Initially, extra demand is met with the cheapest dish, Dish 2. At $D = 47.5$, storage becomes binding. Beyond that point, an extra lunch requires two additional Dish 1 meals and one fewer Dish 2 meal, so its marginal cost is $2(4) - 1 = \$7$.

Maximum number of lunches: 65

Let $N = x_1 + x_2 + x_3$. Storage usage can be written as $0.1N + 0.1x_2 + 0.05x_3$. Because $x_2, x_3 \geq 10$, storage is at least $0.1N + 1.5$. Hence $0.1N + 1.5 \leq 8$ implies $N \leq 65$.

The plan **(45, 10, 10)** attains this bound: storage = 8, cooking time = $175 \leq 180$, and cost = **\$220**. Thus 65 is feasible and is the maximum, including when meals must be whole numbers.

The P1 Demand worksheet contains a formula-based demand grid and an Excel scatter chart. Fractional demand is used only to show the continuous LP sensitivity curve.

Problem 1 | Excel formulas and Solver settings

Solver setting

Specification

Objective: \$B\$12, Min. Changing cells: \$B\$10:\$D\$10.

Constraints: \$B\$15:\$B\$16 $\leq$ \$D\$15:\$D\$16; \$B\$17:\$B\$20 $\geq$ \$D\$17:\$D\$20.

Method: Simplex LP. Make unconstrained variables non-negative. No integer constraints; request Sensitivity report after Solve.

Excel image: P1 Lunch!A23:H26. Solver model is saved in the workbook.

Cell / range	Meaning / formula
B10:D10	Changing cells (35, 15, 10)
B12	=SUMPRODUCT(B5:D5,B10:D10)
B15	=SUMPRODUCT(B6:D6,B10:D10)
B16	=SUMPRODUCT(B7:D7,B10:D10)
B17	=SUM(B10:D10)
B18, B19, B20	=B10 ; =C10 ; =D10
D18, D19, D20	=B8 ; =C8 ; =D8
E15:E16	=D15-B15, filled down
E17:E20	=B17-D17, filled down

Formula documentation exported from Excel

Formula cell(s)	Formula / explanation
B12	=SUMPRODUCT(B5:D5,B10:D10)
B15	=SUMPRODUCT(B6:D6,B10:D10)
E15	=D15-B15
B16	=SUMPRODUCT(B7:D7,B10:D10)
E16	=D16-B16
B17	=SUM(B10:D10)
E17	=B17-D17
B18	=B10
D18	=B8
E18	=B18-D18
B19	=C10
D19	=C8
E19	=B19-D19
B20	=D10
D20	=D8
E20	=B20-D20

Updated Excel image: P1 Lunch!A28:H44. Formulas are written as text so they remain visible alongside the computed model.

Solver objective: minimize B12. Constraints: B15:B16 $\leq$ D15:D16 and B17:B20 $\geq$ D17:D20. Method: Simplex LP, with nonnegative variables. Excel Solver returned status 0 from the starting point (10, 10, 10).

Problem 2 | True / False with justifications

(a) True.

For a continuous LP at an optimal primal-dual solution, complementary slackness gives (shadow price) $\times$ (slack) = 0. A non-binding inequality has strictly positive slack, so its shadow price is zero. This is a statement about LP sensitivity, not integer or nonlinear optimization.

(b) False.

Multiple optima do not require zero allowable coefficient changes for two distinct variables. Counterexample: minimize x_1 , subject to $x_1 \geq 1$ and $0 \leq x_2 \leq 1$. Every $(1, t)$, $0 \leq t \leq 1$, is optimal. The coefficient of x_1 can decrease from 1 to 0 or increase without limit. Only x_2 , whose coefficient is zero, has a zero allowable change in at least one direction. At the optimum $(1, 0)$, its allowable decrease is zero.

(c) False as a guaranteed conclusion.

The two RHS changes occur simultaneously. The 100% rule gives $1/2 + 1/1 = 1.5 = 150\%$, so the separate ranges do not guarantee that the current shadow prices remain valid. The change might be \$8, but the report alone cannot establish it. Re-solve the jointly modified model. Exceeding 100% is inconclusive, not proof that \$8 is impossible.

(d) False.

Resource 1 has allowable increase 2, so its \$5 shadow price is valid only up to RHS 12, holding other data fixed. RHS 14 is outside that range. The objective increase cannot be asserted to equal $5 \times 4 = \$20$.

(e) False as stated.

The \$3 shadow price applies to Resource 2 only over RHS values from $8 - 5 = 3$ through $8 + 1 = 9$, holding everything else fixed. One additional unit, from 8 to 9, changes the objective by \$3. The claim is not valid for every additional unit indefinitely.

(f) True.

Both constraints bind: Resource 1 has final value 10 equal to RHS 10, and Resource 2 has final value 8 equal to RHS 8.

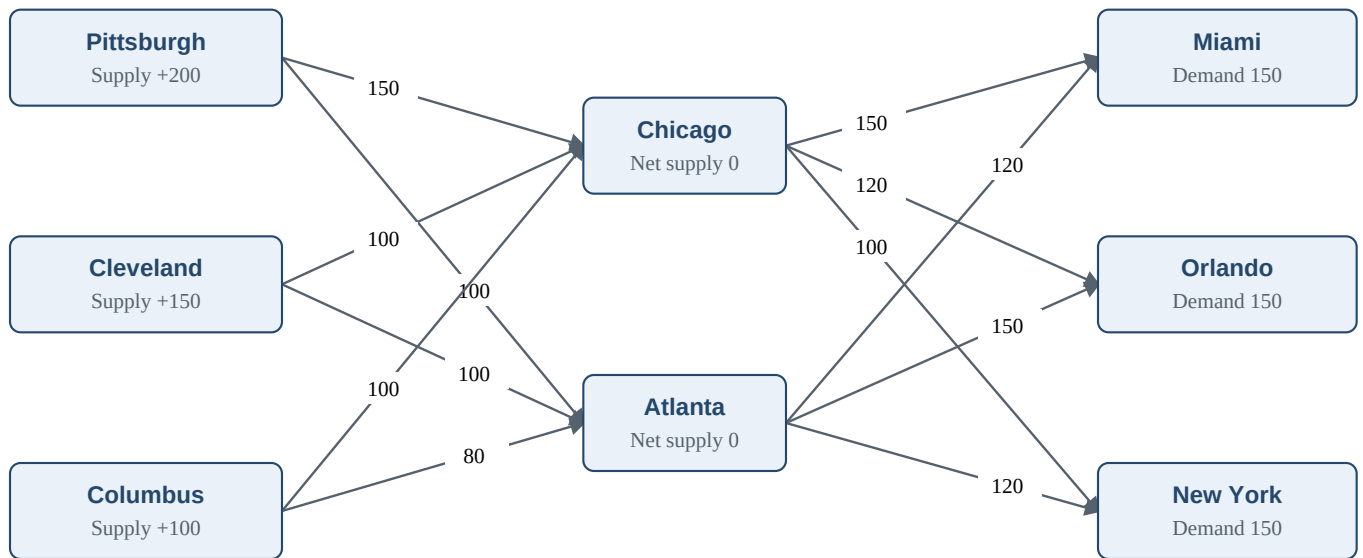
Role of the stated purchase costs

The \$5 and \$4 resource costs do not change the validity ranges. If the objective is profit before paying for newly purchased resources, the net gain from one extra unit is $\$5 - \$5 = \$0$ for Resource 1 and $\$3 - \$4 = -\$1$ for Resource 2, within their individual ranges. Do not subtract purchase costs a second time if they are already included in the stated objective.

Problem 3(a) | Airline network and input data

Origins: Pittsburgh, Cleveland, Columbus. Intermediate hubs: Chicago and Atlanta. Destinations: Miami, Orlando, New York. Total supply and demand are both 450 passengers. The graph shows aggregate node balances; the model also preserves each origin-destination pair.

All arc labels are passenger capacities



OD demand	Miami	Orlando	New York	Origin total
Pittsburgh	100	60	40	200
Cleveland	50	50	50	150
Columbus	0	40	60	100
Destination total	150	150	150	450

Flight arc	Capacity	Cost / passenger	Flight arc	Capacity	Cost / passenger
PIT - CHI	150	\$50	CHI - MIA	150	\$100
PIT - ATL	100	\$100	CHI - ORL	120	\$90
CLE - CHI	100	\$60	CHI - NYC	100	\$80
CLE - ATL	100	\$90	ATL - MIA	120	\$80
COL - CHI	100	\$70	ATL - ORL	150	\$70
COL - ATL	80	\$80	ATL - NYC	120	\$100

Abbreviations: PIT = Pittsburgh; CLE = Cleveland; COL = Columbus; CHI = Chicago; ATL = Atlanta; MIA = Miami; ORL = Orlando; NYC = New York.

Problem 3(b-c) | Network flow formulation and Excel model

(b) Decision variables, parameters, and objective

Let O be the three origins, H the two hubs, and T the three destinations. D_{ij} is demand from origin i to destination j . Let u_{ih} , v_{hj} be flight capacities and a_{ih} , b_{hj} the corresponding per-passenger costs.

Define $y_{ihj} \geq 0$ as passengers from i to j traveling through hub h . These are path flows in the network. One unit of y_{ihj} uses one seat on both $i \rightarrow h$ and $h \rightarrow j$.

Minimize $\sum_{i \in O, h \in H, j \in T} (a_{ih} + b_{hj}) y_{ihj}$.

Constraint	Mathematical statement	Meaning
OD demand	$\sum_h y_{ihj} = D_{ij}$, for every i, j	Preserves all nine OD demands
First-leg capacity	$\sum_j y_{ihj} \leq u_{ih}$, for every i, h	Seats from each origin to each hub
Second-leg capacity	$\sum_i y_{ihj} \leq v_{hj}$, for every h, j	Seats from each hub to each destination
Nonnegativity	$y_{ihj} \geq 0$	No negative passengers

Hub conservation holds automatically: the same y_{ihj} is counted on both legs. Equivalently, use arc flows f_a^{ij} for each OD commodity; outflow minus inflow equals D_{ij} at i , $-D_{ij}$ at j , and 0 elsewhere, with $\sum_{ij} f_a^{ij} \leq \text{capacity}_a$. A model using only aggregate city balances would omit the given OD requirements.

(c) Excel implementation and settings

Each scenario has its own worksheet. Rows 6:14 represent the nine OD pairs. D6:D14 and E6:E14 are flows through Chicago and Atlanta. F6:F14 are direct flows, fixed to zero in the base scenario. All reported solutions are integral.

Solver setting	All P3 worksheets
Objective	Minimize B16 = SUM(L6:L14)
Changing cells	D6:F14
Demand constraints	G6:G14 = C6:C14
Existing flight capacities	E19:E30 $\leq$ C19:C30
Direct flight capacities	D33:D41 $\leq$ C33:C41
Bounds and method	D6:F14 $\geq$ 0; Simplex LP

Row cost L6 = SUMPRODUCT(D6:F6, I6:K6), filled through row 14. Assigned total G6 = SUM(D6:F6). I6 = \$D\$19+\$D\$25 and J6 = \$D\$20+\$D\$28 add the appropriate leg costs. Full formula examples and cell references appear on page 12.

Problem 3(d) | Base network optimum

Minimum total cost: \$69,700 for all 450 passengers. The table below gives the number of passengers assigned to each flight. Multiple routings can achieve the same optimum; this is the solution saved by Excel.

Flight	Passengers	Capacity	Unused seats
PIT - CHI	150	150	0
PIT - ATL	50	100	50
CLE - CHI	100	100	0
CLE - ATL	50	100	50
COL - CHI	20	100	80
COL - ATL	80	80	0
CHI - MIA	50	150	100
CHI - ORL	120	120	0
CHI - NYC	100	100	0
ATL - MIA	100	120	20
ATL - ORL	30	150	120
ATL - NYC	50	120	70

Origin	Destination	Demand	Via Chicago	Via Atlanta	Direct	Assigned	Residual
Pittsburgh	Miami	100	50	50	0	100	0
Pittsburgh	Orlando	60	60	0	0	60	0
Pittsburgh	New York	40	40	0	0	40	0
Cleveland	Miami	50	0	50	0	50	0
Cleveland	Orlando	50	50	0	0	50	0
Cleveland	New York	50	50	0	0	50	0
Columbus	Miami	0	0	0	0	0	0
Columbus	Orlando	40	10	30	0	40	0
Columbus	New York	60	10	50	0	60	0

Excel image: P3 Base!A5:H14. Each row sums to its specified OD demand; the Residual column is zero.

Cost check: first legs cost \$30,800 and second legs cost \$38,900, totaling \$69,700. Exactly 450 seats are used on each connecting leg across the network.

Problem 3(e-f) | Add direct flights from Pittsburgh

(e) Prediction before optimization

A reasonable initial prediction is **Pittsburgh**, because it supplies 200 passengers, the largest origin demand. Its three direct flights can accommodate all 200 and avoid expensive connecting itineraries. This is only a heuristic: the best choice also depends on which constrained hub seats are freed for the other origins.

Model changes for each direct-flight scenario

Add $z_{ij} \geq 0$ for direct passengers and add $\sum_{ij} d_{ij} z_{ij}$ to the objective. Replace each demand equation with $\sum_h y_{ihj} + z_{ij} = D_{ij}$. For selected origin k , enforce $z_{kj} \leq \text{direct capacity}_{kj}$; for all $i \neq k$ enforce $z_{ij} = 0$. Existing connecting-flight capacity constraints stay unchanged because direct passengers do not use those arcs.

(f) Pittsburgh direct flights

Direct flight	Capacity	Unit cost	Optimal passengers
PIT - MIA	100	\$120	100
PIT - ORL	60	\$80	60
PIT - NYC	40	\$70	40

New bounds: $z_{\text{PIT,MIA}} \leq 100$, $z_{\text{PIT,ORL}} \leq 60$, $z_{\text{PIT,NYC}} \leq 40$; all other direct flows are zero. In P3 Pittsburgh, C33:C35 contain these limits and C36:C41 are zero.

Minimum cost: \$57,900; savings: \$11,800. All 200 Pittsburgh passengers travel directly. Direct-flight cost is \$19,600; the other 250 passengers connect at a cost of \$38,300.

Origin	Destination	Demand	Via Chicago	Via Atlanta	Direct	Assigned	Residual
Pittsburgh	Miami	100	0	0	0	100	100
Pittsburgh	Orlando	60	0	0	0	60	60
Pittsburgh	New York	40	0	0	0	40	40
Cleveland	Miami	50	0	0	50	0	50
Cleveland	Orlando	50	50	0	0	0	50
Cleveland	New York	50	50	0	0	0	50
Columbus	Miami	0	0	0	0	0	0
Columbus	Orlando	40	0	0	40	0	40
Columbus	New York	60	50	10	0	0	60

Excel image: P3 Pittsburgh!A5:H14. Direct decisions are in F6:F8. All demand residuals are zero.

The full 12-arc connecting-flight assignment for this scenario is included in the comparison table on page 11.

Problem 3(g-h) | Cleveland and Columbus alternatives

(g) Add direct flights from Cleveland

Direct flight	Capacity	Unit cost	Passengers
CLE - MIA	100	\$110	50
CLE - ORL	60	\$75	50
CLE - NYC	50	\$65	50

Set $z_{\text{CLE,MIA}} \leq 100$, $z_{\text{CLE,ORL}} \leq 60$, $z_{\text{CLE,NYC}} \leq 50$; all other direct flows are zero. In P3 Cleveland, C36:C38 contain these limits. Use the changed OD equations on page 9 and retain every connecting capacity.

Minimum cost: \$57,600; savings: \$12,100. All 150 Cleveland passengers fly directly, costing \$12,500. The remaining 300 connect at a cost of \$45,100.

Origin	Destination	Demand	Via Chicago	Via Atlanta	Direct	Assigned	Residual
Pittsburgh	Miami	100	50	50	0	100	0
Pittsburgh	Orlando	60	60	0	0	60	0
Pittsburgh	New York	40	40	0	0	40	0
Cleveland	Miami	50	0	0	50	50	0
Cleveland	Orlando	50	0	0	50	50	0
Cleveland	New York	50	0	0	50	50	0
Columbus	Miami	0	0	0	0	0	0
Columbus	Orlando	40	0	40	0	40	0
Columbus	New York	60	60	0	0	60	0

Excel image: P3 Cleveland!A5:H14. Direct decisions are F9:F11.

(h) Add direct flights from Columbus

Direct flight	Capacity	Unit cost	Passengers
COL - MIA	80	\$100	0
COL - ORL	40	\$70	40
COL - NYC	60	\$60	60

Set $z_{\text{COL,MIA}} \leq 80$, $z_{\text{COL,ORL}} \leq 40$, $z_{\text{COL,NYC}} \leq 60$; all other direct flows are zero. In P3 Columbus, C39:C41 contain these limits. Columbus-Miami demand is zero, so this new flight carries no one.

Minimum cost: \$59,500; savings: \$10,200. All 100 Columbus passengers fly directly, costing \$6,400. The other 350 connect at a cost of \$53,100.

Origin	Destination	Demand	Via Chicago	Via Atlanta	Direct	Assigned	Residual
Pittsburgh	Miami	100	50	50	0	100	0
Pittsburgh	Orlando	60	60	0	0	60	0
Pittsburgh	New York	40	40	0	0	40	0
Cleveland	Miami	50	0	50	0	50	0
Cleveland	Orlando	50	50	0	0	50	0
Cleveland	New York	50	50	0	0	50	0
Columbus	Miami	0	0	0	0	0	0
Columbus	Orlando	40	0	0	40	40	0
Columbus	New York	60	0	0	60	60	0

Excel image: P3 Columbus!A5:H14. Direct decisions are F12:F14.

Problem 3(f-i) | Flight assignments and recommendation

All entries are passengers. Capacities refer to the original connecting flights. These assignments, together with the direct-flight counts on pages 9-10, specify every flight in each scenario.

Connecting flight	Capacity	Base	PIT direct	CLE direct	COL direct
PIT - CHI	150	150	0	150	150
PIT - ATL	100	50	0	50	50
CLE - CHI	100	100	100	0	100
CLE - ATL	100	50	50	0	50
COL - CHI	100	20	50	60	0
COL - ATL	80	80	50	40	0
CHI - MIA	150	50	0	50	50
CHI - ORL	120	120	50	60	110
CHI - NYC	100	100	100	100	90
ATL - MIA	120	100	50	50	100
ATL - ORL	150	30	40	40	0
ATL - NYC	120	50	10	0	0

Scenario	Total cost	Savings vs. base	Direct passengers
Base	\$69,700	\$0	0
Pittsburgh	\$57,900	\$11,800	200
Cleveland	\$57,600	\$12,100	150
Columbus	\$59,500	\$10,200	100

(i) Recommendation: Cleveland

Offer the direct flights from Cleveland. Its total cost of \$57,600 is the lowest, saving \$12,100 relative to the base network, or approximately 17.36%. It is \$300 cheaper than Pittsburgh direct flights and \$1,900 cheaper than Columbus direct flights.

Cleveland wins even though Pittsburgh has more passengers. The total-cost comparison includes both the direct fares and the best rerouting of all remaining connecting passengers. The initial largest-demand heuristic therefore does not determine the best network-wide decision.

This recommendation uses the supplied per-passenger costs only. The problem does not give fixed launch costs or require every new flight to carry passengers; neither has been added to the model.

Problem 3 | Spreadsheet formula documentation

Solver setting **Specification**

Objective \$B\$16, Min; changing cells \$D\$6:\$F\$14; Simplex LP; non-negative variables.

Constraints: \$G\$6:\$G\$14 = \$C\$6:\$C\$14; \$E\$19:\$E\$30 <= \$C\$19:\$C\$30; \$D\$33:\$D\$41 <= \$C\$33:\$C\$41.

Connecting path costs = first-leg cost + second-leg cost. Flow formulas aggregate all OD passengers using each flight. Capacities and OD demands stay fixed across scenarios.

Formula cell(s)	Formula / explanation
G6 (fill to G14)	=SUM(D6:F6)
H6 (fill to H14)	=G6-C6
I6, J6	=\$D\$19+\$D\$25 ; =\$D\$20+\$D\$28
L6 (fill to L14)	=SUMPRODUCT(D6:F6,I6:K6)
B16	=SUM(L6:L14)
E19, E20	=SUM(D6:D8) ; =SUM(E6:E8)
E25	=D6+D9+D12
F19 (fill to F30)	=C19-E19
D33, E33	=F6 ; =C33-D33

Excel image: P3 Base!A43:L57. The same formulas and Solver constraints apply to all four scenario worksheets.

Cell(s)	Formula / mapping
D6:F14	Decision variables: Chicago, Atlanta, direct
G6, H6	=SUM(D6:F6) ; =G6-C6 (fill down)
I6, J6	=\$D\$19+\$D\$25 ; =\$D\$20+\$D\$28
L6	=SUMPRODUCT(D6:F6,I6:K6) (fill down)
B16	=SUM(L6:L14)
E19, E20	=SUM(D6:D8) ; =SUM(E6:E8)
E25	=D6+D9+D12
F19	=C19-E19 (capacity slack; fill down)
D33, E33	=F6 ; =C33-D33 (direct flow / slack)

First-leg rows 19:24 are ordered by origin, then Chicago/Atlanta. Second-leg rows 25:30 are Chicago to the three destinations, then Atlanta to the three destinations. Direct rows 33:41 follow the same OD order as rows 6:14. This mapping makes every flight capacity and each path cost auditable.

Excel Solver returned status 0 for all four scenarios from zero-valued decision cells. All OD residuals are zero, every capacity slack is nonnegative, and every final decision is an integer.

Problem 4(a-b) | Mentoring network model

(a) Parameters and decision variables

Create a mentor copy and a mentee copy of every person. Let E be the allowed mentor-mentee pairs with a numeric preference score in the supplied table; s_{ij} is that score. A dash is forbidden, including self-mentoring. These allowed pairs encode the stated role restrictions.

Let $x_{ij} = 1$ if mentor i is assigned to mentee j and 0 otherwise, for (i,j) in E . **Maximize** $\sum_{(i,j) \in E} s_{ij} x_{ij}$, subject to:

$\sum_{i:(i,j) \in E} x_{ij} = 1$ for every mentee j ;

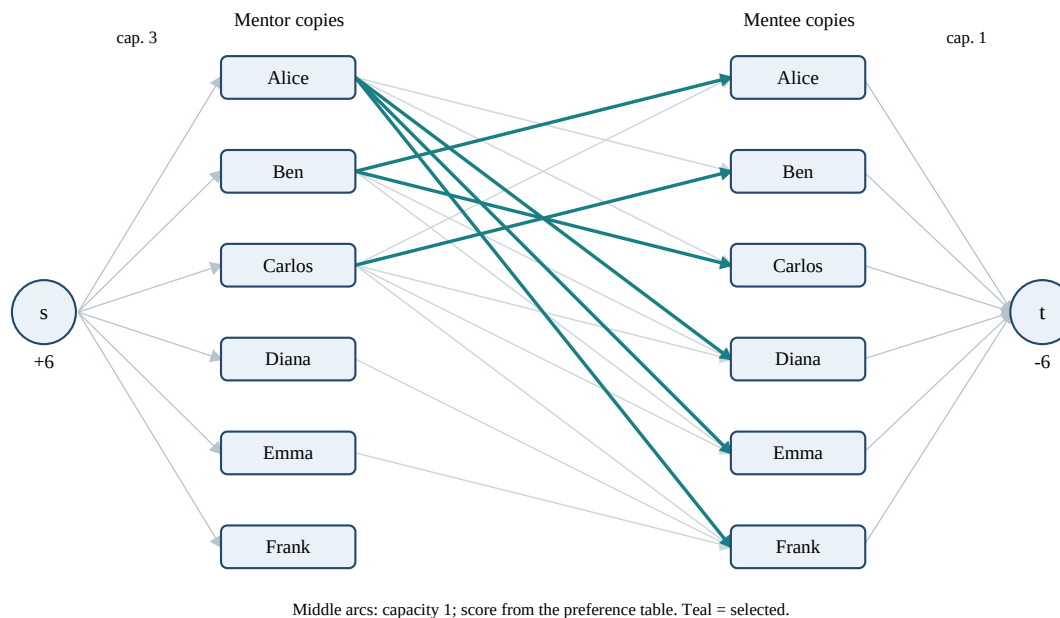
$\sum_{j:(i,j) \in E} x_{ij} \leq 3$ for every mentor i ;

$0 \leq x_{ij} \leq 1$ on allowed pairs; $x_{ij} = 0$ on forbidden pairs.

The network construction is source $s \rightarrow$ mentor i (capacity 3, cost 0), mentor $i \rightarrow$ mentee j on allowed pairs (capacity 1, cost $-s_{ij}$), and mentee $j \rightarrow$ sink t (capacity 1, cost 0). Send exactly six units and minimize total cost. Source supply is +6, sink supply is -6, and all other nodes have net supply 0.

All six mentee-to-sink arcs must be saturated, so every person gets exactly one mentor. Integral capacities make the network LP admit an integral optimum; binary constraints are optional for this formulation.

(b) Graphical representation



A separate copy on each side lets a person be both mentor and mentee. Grey middle arcs are eligible but unselected; teal middle arcs show the optimal assignment. Frank has no outgoing mentor arcs.

Mentor / mentee	Alice	Ben	Carlos	Diana	Emma	Frank
Alice	-	5	10	100	95	85
Ben	65	-	20	45	20	-
Carlos	10	70	-	60	40	25
Diana	-	-	-	-	-	65
Emma	-	-	-	-	-	50
Frank	-	-	-	-	-	-

Problem 4(c-d) | Excel solution and company results

(c) Excel Solver setup

Maximize P4 Mentoring!B33 = SUMPRODUCT(B5:G10,B15:G20), changing B15:G20. Enforce H15:H20 $\leq$ I15:I20 (mentor capacities), B22:G22 = B23:G23 (one mentor each), B15:G20 $\leq$ B26:G31 (allowed-pair mask), and nonnegativity. Use Simplex LP. Solver returned status 0 with an integral solution from an all-zero starting point.

Assign: mentor / mentee	Alice	Ben	Carlos	Diana	Emma	Frank	Mentee load	Capacity
Alice	0	0	0	1	1	1	3	3
Ben	1	0	1	0	0	0	2	3
Carlos	0	1	0	0	0	0	1	3
Diana	0	0	0	0	0	0	0	3
Emma	0	0	0	0	0	0	0	3
Frank	0	0	0	0	0	0	0	3
Mentors assigned	1	1	1	1	1	1		
Required	1	1	1	1	1	1		

Excel image: P4 Mentoring!A14:I23. Rows are mentors and columns are mentees; 1 means assigned. H15:H20 show mentor loads.

(d) Assignments to report to the company

Mentee	Assigned mentor	Preference score
Alice	Ben	65
Ben	Carlos	70
Carlos	Ben	20
Diana	Alice	100
Emma	Alice	95
Frank	Alice	85
Total		435

Total preference score: 435. Alice mentors three people, Ben mentors two, Carlos mentors one, and Diana, Emma, and Frank mentor none. Every person has exactly one mentor, every pairing is allowed, and no mentor exceeds three mentees.

Optimality check: the maximum available scores for the six individual mentees are 65, 70, 20, 100, 95, and 85. Their sum is 435. This assignment attains all six individual maxima while respecting mentor capacities, so no feasible assignment can do better.

Problem 4(e) | Interpretation, limitations, and improvements

Within the supplied scores and eligibility rules, **everyone receives their highest-scoring eligible mentor**. The solution is therefore optimal both in total score and for each person individually among the listed options. This still does not guarantee a strong real-world mentoring program.

Preference scores and limited options

Carlos receives a score of only 20, but that is his best allowed match: Alice scores 10 and Ben scores 20. Reassigning the existing people cannot improve Carlos beyond 20. HR could recruit additional eligible mentors, gather more candidate matches, and ask whether the scores are comparable across people. Minimum acceptable-score constraints or a max-min objective can protect satisfaction, but a threshold above 20 makes this particular candidate set infeasible.

Reciprocal mentoring and hierarchy

Ben mentors Carlos and Carlos mentors Ben. That may weaken independence or create a conflict in feedback. If reciprocal pairs are undesirable, add $x_{ij} + x_{ji} \leq 1$ for eligible pairs. However, requiring every person in the highest role to have a same-level mentor necessarily creates some directed cycle within that finite highest-role group. Prohibiting every cycle while retaining those rules would make the model infeasible. Allow an external senior mentor or exempt selected highest-role staff to remove that structural issue.

Workload and suitability beyond one score

Three mentees need not require equal effort. Replace the uniform cap with mentor-specific available hours and estimated mentoring time, and include availability, skill coverage, conflicts of interest, and voluntary participation. Collect feedback after a trial period and update the scores before rerunning the model.

Excel formula documentation

Solver setting	Specification
Objective \$B\$33, Max; changing cells \$B\$15:\$G\$20; Simplex LP; non-negative variables.	
Constraints: \$H\$15:\$H\$20 <= \$I\$15:\$I\$20; \$B\$22:\$G\$22 = \$B\$23:\$G\$23; \$B\$15:\$G\$20 <= \$B\$26:\$G\$31.	
This network LP has an integral optimum. Binary constraints may alternatively be added to B15:G20; the optimum remains 435.	

Formula cell(s)	Formula / explanation
H15 (fill to H20)	=SUM(B15:G15)
B22 (fill across to G22)	=SUM(B15:B20)
B33	=SUMPRODUCT(B5:G10,B15:G20)

Excel image: P4 Mentoring!A35:J43. The workbook records exact constraints and readable objective/load formulas.

Source documents: HW2.pdf, Problems 1-4; Submission example.pdf, required submission components. The submitted workbook is DMU_HW2_mnagersh.xlsx. Models were solved in Microsoft Excel Solver and independently verified with SciPy/HiGHS. The mentoring optimum was also checked by exhaustive enumeration.

Problem 1 | Excel Solver Parameters

P1 Lunch: minimize B12 by changing B10:D10. The screenshot shows the saved cooking, storage, demand, and variety constraints.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$12'. The 'To' section has 'Min' selected, and the 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$B\$10:\$D\$10'. The 'Subject to the Constraints' list contains two constraints: '\$B\$15:\$B\$16 <= \$D\$15:\$D\$16' and '\$B\$17:\$B\$20 >= \$D\$17:\$D\$20'. The 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method' dropdown is set to 'Simplex LP'. The 'Options' button is visible. At the bottom, there are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

\$B\$15:\$B\$16 <= \$D\$15:\$D\$16	Add Change Delete Reset All Load/Save
\$B\$17:\$B\$20 >= \$D\$17:\$D\$20	

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P1 Lunch.

Problem 3(c) | Base model Solver Parameters

P3 Base: minimize B16 by changing D6:F14. The direct-flight capacity cells C33:C41 are all zero in this scenario.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$16'. The 'To' section has three radio buttons: 'Max' (unselected), 'Min' (selected), and 'Value Of' (unselected). The 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$D\$6:\$F\$14'. The 'Subject to the Constraints' list contains four constraints: '\$D\$33:\$D\$41 <= \$C\$33:\$C\$41', '\$D\$6:\$F\$14 >= 0', '\$E\$19:\$E\$30 <= \$C\$19:\$C\$30', and '\$G\$6:\$G\$14 = \$C\$6:\$C\$14'. To the right of the constraints list are five buttons: 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints list is a checked checkbox labeled 'Make Unconstrained Variables Non-Negative'. The 'Select a Solving Method' dropdown is set to 'Simplex LP', and there is an 'Options' button next to it. At the bottom of the dialog is a 'Solving Method' section with a text box explaining the different engines: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.' At the very bottom are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P3 Base.

Problem 3(f) | Pittsburgh Solver Parameters

P3 Pittsburgh: minimize B16 by changing D6:F14. C33:C35 allow Pittsburgh direct flights; all other direct-flight capacities are zero.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$16'. The 'To' section has radio buttons for 'Max', 'Min' (selected), and 'Value Of:'. The 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$D\$6:\$F\$14'. The 'Subject to the Constraints' list contains four constraints: '\$D\$33:\$D\$41 <= \$C\$33:\$C\$41', '\$D\$6:\$F\$14 >= 0', '\$E\$19:\$E\$30 <= \$C\$19:\$C\$30', and '\$G\$6:\$G\$14 = \$C\$6:\$C\$14'. To the right of the list are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints list is a checked checkbox for 'Make Unconstrained Variables Non-Negative'. The 'Select a Solving Method' dropdown is set to 'Simplex LP', with an 'Options' button to its right. A 'Solving Method' section at the bottom provides instructions on selecting the GRG Nonlinear, LP Simplex, or Evolutionary engine. At the bottom of the dialog are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P3 Pittsburgh.

Problem 3(g) | Cleveland Solver Parameters

P3 Cleveland: minimize B16 by changing D6:F14. C36:C38 allow Cleveland direct flights; all other direct-flight capacities are zero.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$16'. The 'To' section has three radio buttons: 'Max' (unselected), 'Min' (selected), and 'Value Of' (unselected). The 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$D\$6:\$F\$14'. The 'Subject to the Constraints' list contains four constraints: '\$D\$33:\$D\$41 <= \$C\$33:\$C\$41', '\$D\$6:\$F\$14 >= 0', '\$E\$19:\$E\$30 <= \$C\$19:\$C\$30', and '\$G\$6:\$G\$14 = \$C\$6:\$C\$14'. To the right of the constraints list are five buttons: 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints list is a checkbox labeled 'Make Unconstrained Variables Non-Negative' which is checked. The 'Select a Solving Method' dropdown is set to 'Simplex LP'. To the right of the dropdown is an 'Options' button. Below the 'Options' button is a box titled 'Solving Method' with the text: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.' At the bottom of the dialog are two buttons: 'Close' and 'Solve'.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P3 Cleveland.

Problem 3(h) | Columbus Solver Parameters

P3 Columbus: minimize B16 by changing D6:F14. C39:C41 allow Columbus direct flights; all other direct-flight capacities are zero.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$16'. The 'To' section has three radio buttons: 'Max', 'Min' (which is selected), and 'Value Of:'. The 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$D\$6:\$F\$14'. The 'Subject to the Constraints' list contains four constraints: '\$D\$33:\$D\$41 <= \$C\$33:\$C\$41', '\$D\$6:\$F\$14 >= 0', '\$E\$19:\$E\$30 <= \$C\$19:\$C\$30', and '\$G\$6:\$G\$14 = \$C\$6:\$C\$14'. To the right of this list are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints list is a checked checkbox labeled 'Make Unconstrained Variables Non-Negative'. The 'Select a Solving Method' dropdown is set to 'Simplex LP', and there is an 'Options' button next to it. At the bottom, there is a 'Solving Method' section with explanatory text: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.' At the very bottom are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P3 Columbus.

Problem 4(c) | Mentoring Solver Parameters

P4 Mentoring: maximize B33 by changing B15:G20. Constraints enforce mentor capacities, one mentor per mentee, allowed matches, and nonnegativity.

The screenshot shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective' field is set to '\$B\$33'. The 'To' section has 'Max' selected, and the 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$B\$15:\$G\$20'. The 'Subject to the Constraints' list contains four constraints: '\$B\$15:\$G\$20 <= \$B\$26:\$G\$31', '\$B\$15:\$G\$20 >= 0', '\$B\$22:\$G\$22 = \$B\$23:\$G\$23', and '\$H\$15:\$H\$20 <= \$I\$15:\$I\$20'. To the right of this list are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints, the 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method' dropdown is set to 'Simplex LP', with an 'Options' button to its right. A 'Solving Method' section at the bottom provides instructions on selecting the GRG Nonlinear, LP Simplex, or Evolutionary engine. At the very bottom are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Actual screenshot of the Excel Solver Parameters dialog for P4 Mentoring.